INCH-POUND

MIL-O-44278A
1 May 1989
SUPERSEDING
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7 August 1987

MILITARY SPECIFICATION

OATMEAL, INSTANT (OPERATIONAL RATION COMPONENT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers instant oatmeal for use by the Department of Defense as a component of Ration, Cold Weather.

1.2 Classification. The instant oatmeal shall be the following types as specified (see 6.1).

Type I - Apple and cinnamon
Type II - Maple and brown sugar
Type III - Strawberry and cream

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

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SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	- Aluminum Foil
PPP-B-636	- Boxes, Shipping, Fiberboard

STANDARDS

FEDERAL

FED-STD-595	- Colors

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Dehydrated (Low Moisture) Apples

(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Washington, DC 20090-6456)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique
F 88 - Standard Test Method for Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists Inc., 3340 Pilot Knob Road, St. Paul, MN 55121.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1120, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to the first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign materials, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Rolled oats, instant, (groats). Instant rolled cooking oats shall be thin, uniform flakes produced by cleaning, dehulling, drying, steaming, cutting, and rolling clean, sound, quality whole or steel cut oat groats. The product shall be 100 percent natural material with no additives, and shall be heated to inactivate enzymes. The oats shall test negative for tyrosinase (catechol oxidase). The oats shall have a bright creamy or tan color. Rolled oats are considered instant when they can be reconstituted in boiling water in approximately 1 minute. The rolled oats shall meet the following requirements:

Moisture - not more than 10.5 percent
Flake thickness - not more than 0.015 inches
Oat hulls - 1 per ounce maximum
Not more than 4 percent shall remain on U.S. Standard No. 4 sieve
Not more than 70 percent shall pass through a U.S. Standard No. 8 sieve

3.2.2 Creamer, nondairy, dry. The dry nondairy creamer shall be a white to light cream color, free-flowing, uniform, granular powder, free from lumps. The creamer shall be free from foreign material and shall not contain noticeable brown and/or black scorched particles. When rehydrated, the creamer shall readily dissolve and impart a sweet creamy flavor. The creamer shall contain corn syrup solids, partially hydrogenated vegetable oils and other safe and suitable ingredients. The creamer shall contain not more than 4.9 percent moisture, not less than 31 percent fat, and not less than 95 percent shall pass through a U.S. No. 10 sieve.

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3.2.3 Guar gum. Guar gum shall comply with the requirements of the Food Chemicals Codex and shall not exceed 15 percent moisture. A 1 percent solution of guar gum shall have a minimum 3000 cps viscosity. Not more than 5 percent of the guar gum shall remain on a U.S. Standard No. 100 sieve and not less than 75 percent shall pass through a U.S. Standard No. 200 sieve (see 6.3).

3.2.4 Salt. The salt shall be clean, white, refined sodium chloride with or without anti-caking agents and shall be a flour salt granulation.

3.2.5 Sugar, white, granulated. Sugar shall be white refined, granulated cane or beet sugar, or a combination thereof. The sugar shall comply with industry bulk fine or ready mix grades.

3.2.6 Sugar, brown, granulated. Granulated brown sugar shall be a granular, free-flowing sugar with natural brown sugar flavor. The granules shall be aggregates which break apart in solution, yielding a dark golden brown liquid. Particle size pattern on U.S. Standard sieves shall be not less than 6 percent remaining on a No. 16 mesh and less than 8 percent passing through a No. 50 mesh screen. The product shall be resistant to caking and have a maximum moisture of 1.0 percent.

3.2.7 Cinnamon, ground. Ground cinnamon shall be prepared from the dried bark of Cinnamomum burmanii Blume commonly known as "Korintz Cinnamon". The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.8 Apple flakes, low moisture. The low moisture apple flakes shall be prepared from clean, sound, mature, peeled and cored apples. The low moisture (dehydrated) apple flakes shall be Grade B or better of the US Standards for Grades of Dehydrated (Low Moisture) Apples, except flake thickness shall be 0.023 to 0.050 inches. The flavor shall be a typical sweet-tart characteristic of apples and free from off-flavors. Moisture content shall not exceed 3.5 percent. Food grade calcium stearate may be used as an anti-caking agent. The use of FDA approved color retention agents is permitted. (See 6.4.)

3.2.9 Apple flakes, low moisture, strawberry flavored and colored. The low moisture apple flakes shall be prepared from clean, sound, mature, peeled and cored apples. The low moisture (dehydrated) apple flakes shall be Grade B or better of the US Standards for Grades of Dehydrated (Low Moisture) Apples, except flake thickness shall be 0.023 to 0.065 inches. The apple flakes shall be flavored and colored with a strawberry juice solids concentrate containing artificial flavoring and coloring agents, citric acid, and food grade calcium stearate. The use of FDA approved color retention agents is permitted. The strawberry flavored and colored apple flakes shall be dry and free-flowing, possessing a bright red to reddish pink color typical of strawberry. The strawberry flavored and colored apple flakes shall possess a slightly sweet-tart flavor and be of an aroma characteristic of strawberries. Moisture content shall not exceed 3.5 percent. (See 6.4).

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3.2.10 Flavoring, maple, natural, spray dried. Natural maple flavor shall be equal to Spray Dried Natural Maple Flavor WONF #24029 manufactured by Food Materials Corporation.

3.2.11 Flavoring, apple, natural and artificial. Natural and artificial apple flavor shall be equal to Flavor-Cap Natural and Artificial Apple Flavor #F22311 manufactured by McCormick-Stange Flavor Company, Inc. (See 6.5).

3.2.12 Flavoring, strawberry, artificial. Artificial strawberry flavor shall be equal to Caps Artificial Strawberry Flavor No. 1979 manufactured by Borden (See 6.6.)

3.3 Preparation of types of instant oatmeal.

3.3.1 Preparation of apple and cinnamon oatmeal.

3.3.1.1 Formulation of apple and cinnamon oatmeal. The apple and cinnamon oatmeal shall be formulated as follows:

Ingredients	Percent by weight
Oats, rolled, instant	50.8
Creamer, nondairy, dry	20.0
Sugar, white, granulated	19.45
Apple flakes, low moisture	8.0
Guar gum	1.0
Cinnamon	0.4
Salt	0.3
Flavoring, apple, natural and artificial	0.05

3.3.1.2 Mixing of apple and cinnamon oatmeal. Mix all ingredients except for rolled oats and apple flakes in a ribbon blender or other suitable mixer until homogenous. Blend rolled oats and apple flakes with blended dry ingredients with suitable equipment such that requirements of 3.4e and 3.4h are met.

3.3.2 Preparation of maple and brown sugar oatmeal.

3.3.2.1 Formulation of maple and brown sugar oatmeal. The maple and brown sugar oatmeal shall be formulated as follows:

Ingredients	Percent by weight
Oats, rolled, instant	53.5
Brown sugar, granulated	25.0
Creamer, nondairy, dry	20.0
Guar gum	1.0
Salt	0.3
Flavoring, maple, natural, spray dried	0.2

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3.3.2.2 Mixing of maple and brown sugar oatmeal. Mix all ingredients except for rolled oats in a ribbon blender or other suitable mixer until homogenous. Blend rolled oats with blended dry ingredients with suitable equipment such that requirements of 3.4e and 3.4h are met.

3.3.3. Preparation of strawberry and cream oatmeal.

3.3.3.1 Formulation of strawberry and cream oatmeal. The strawberry and cream oatmeal shall be formulated as follows:

Ingredients	Percent by weight
Oats, rolled, instant	46.5
Creamer, nondairy, dry	25.0
Sugar, white, granulated	19.0
Apple flakes, strawberry flavored and colored	8.0
Guar gum	1.0
Salt	0.3
Flavoring, strawberry, artificial	0.2

3.3.3.2 Mixing of strawberry and cream oatmeal. Mix all ingredients except for rolled oats and apple flakes in a ribbon blender or other suitable mixer until homogenous. Blend rolled oats and apple flakes with blended dry ingredients with suitable equipment such that requirements of 3.4e and 3.4h are met.

3.4 Finished product. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. The contents of each pouch shall weigh not less than 125.0 grams and not more than 135.0 grams.

e. The apple and cinnamon oatmeal shall be light beige rolled oat flakes with light cream-yellow apple flakes, cream colored powder, and dark cinnamon particles. The maple and brown sugar oatmeal shall be light beige rolled oat flakes with small tan sugar particles and cream colored powder. The strawberry and cream oatmeal shall be light beige rolled oat flakes with bright reddish pink apple flakes and cream colored powder.

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f. Moisture content of the dry mix shall not exceed 7.5 percent.

g. When product is rehydrated as specified in footnote 3 of table III, the product shall possess flavor, odor and texture, similar to that of rehydrated commercial instant oatmeal.

h. The rehydrated product shall possess distinct oat flakes and (if applicable) apple flakes.

3.4.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.6 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identify, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.1.2 Material certification. Material listed below shall be accepted on the basis of a contractor's certificate of conformance to the indicated requirements.

Material requirement	Requirement paragraph	Test procedure
Thickness of films for laminated pouches and for mixing bags	5.1.1.1 5.1.1.1.1	L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378
Aluminum foil thickness	5.1.1.1.1	As specified in QQ-A-1876
Laminated material identification and construction	5.1.1.1.1 5.1.1.3	Laboratory evaluation
Mixing bag material identification	5.1.1.1	Laboratory evaluation
Color of laminated material.	5.1.1.1.1	Visual evaluation

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4.5.2 Unfilled mixing bag examination. The unfilled mixing bags shall be examined for the defects listed in table I. The lot size shall be expressed in bags. The sample unit shall be one unfilled mixing bag. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65 for major defects.

TABLE I. Unfilled mixing bag defects

Category	Defect
Major	
101	Not formed as specified
102	Inside demensions not as specified
103	Tear, hole, or open seal
104	Not material specified
105	Bags stick together
106	Not clean 1/
107	Foreign odor

1/ The bag shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal fillings, etc.) or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean, dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one mixing bag face (localized and aggregate).

c. Water spots.

4.5.3 Unfilled mixing bag leakage testing. The unfilled mixing bags shall be tested for leakage as follows: Each sample bag shall be filled with 8 ounces of boiling water (being careful not to spill any water on the outside) and suspended 8 to 12 inches above a sheet of clean paper for 10 minutes. Any moisture on the paper attributable to bag leakage shall be classified as a major defect. The lot size shall be expressed in bags. The sample unit shall be one bag. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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4.5.4 Unfilled preformed pouch seal strength testing. The unfilled pouches shall be tested for seal strength in accordance with ASTM Standard Test Method F 88 except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut 1/2 or 1 inch in width and to a length suitable for proper mounting. The lot size shall be expressed in pouches. The sample size shall be the number of pouches indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen or average seal strength failing to meet the requirements specified in 5.1.1.3 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.5 In-process examination. In-process examination shall be performed to determine conformance to formulation, processing, pouch filling, pouch sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of formulation or other valid documents shall be cause for rejection of the involved product.

4.5.6 Examination for excess air in pouch. The contents of one filled and sealed pouch shall be evenly distributed in the pouch and the pouch shall be placed on a flat surface. A straight edge ruler shall be placed across one face of the pouch at its highest point. The distance between the top of the flat surface and the bottom of the straight edge shall be measured at each end of the straight edge. The two measurements shall be averaged. An average thickness exceeding 7/8 inch shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one pouch. The inspection level shall be S-2 and the AQL expressed in terms of defects per hundred units shall be 1.5.

4.5.7 Filled and sealed pouch examination. The filled and sealed pouches shall be examined for the defects listed in table II. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE II. Filled and sealed pouch defects 1/

Category		Defect
Major	Minor	
101		Tear, hole, or open seal
102		Not material specified
103		Pouch dimensions not as specified
104		Bottom and side seams not heat sealed
105		Width of bottom and side seals not as specified

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TABLE II. Filled and sealed pouch defects 1/ (cont'd)

Category		Defect
Major	Minor	
106		Distance between inside edge of tear notch and inside edge of seal is less than 3/16 inch
107		Exterior color of pouch is not as specified
108		Pouch has odor not associated with material
109		Not clean 2/
110		Closure of top seal extends into or below tear nick or notch locations
111		Closure seal not produced by means of heat
112		Closure less than specified width
113		Closure seal not located as specified
114		Presence of delamination 3/
115		Required labeling missing, incorrect, illegible, or smudged
116		Presence of entrapped matter (for example, loose oats, powders, granules, mixing bag material, etc.) that reduces the effective closure to less than 1/16 inch 4/
117		Fold over wrinkles in closure seal
	201	Tear nick or notch missing
	202	Tear nick or notch not located as specified
	203	Depth of tear nick or notch not as specified
	204	Serrations missing 5/
	205	Serrations not located as specified 5/
	206	Depth of serrations not as specified 5/

1/ Evidence of rodent or insect infestation shall be cause for rejection of the lot.

2/ The pouch shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal, filings, etc.) or generally detract from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Dried product which affects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

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3/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Pouches exhibiting this type of delamination shall be tested by manually flexing the delamination area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall be rapidly flexed by rotating both hands in alternating clockwise-counter-clockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, the pouch shall be rejected.

4/ The effective closure seal is defined as any uncontaminated, fusion bonded, continuous path, minimum 1/16 inch wide from side seal to side seal that produces a hermetically sealed pouch.

5/ Applicable only to pouches made on machinery that provides serrations.

4.5.8 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty mixing bag and empty pouch. Results shall be reported to the nearest 1 gram. Any individual net weight of less than 125.0 grams or more than 135.0 grams shall be scored as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5.9 Product inspection. The finished product shall be examined for the defects listed in table III. The lot size shall be expressed in pouches. The sample unit shall be the contents of one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects.

TABLE III. Product defects 1/ 2/

Category	Defect
Major	
101	Color of the applicable type of instant oatmeal not as specified.

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TABLE III. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	
	Applicable to rehydrated product 3/
102	Rehydrated product does not possess flavor, odor and texture similar to rehydrated commercial instant oatmeal.
103	When product is rehydrated, it does not possess distinct oat flakes and, if applicable, apple flakes.

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot.

3/ Add 8 fluid ounces of boiling water to a mixing bag containing product. Knead to mix and let sit 2 minutes. Up to 2 additional ounces of boiling water may be added for typical consistency.

4.5.10 Moisture content testing. The moisture content of the instant oatmeal shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists (AOAC): Chapter: Cereal Foods; Section: Total Solids (Moisture, Indirect Method); Method: Vacuum Oven Method. Results shall be reported to the nearest 0.1 percent. Any result not conforming to the moisture content requirement in 3.4f shall be classified as a major defect. The lot size shall be expressed in units of pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.11 Pouch leakage, delamination and seal separation testing. The filled and sealed pouches shall be tested by placing them in a dry desicator, or similar apparatus, and subjecting them to a vacuum of 26 inches of mercury (atmospheric pressure is 29.9 inches of mercury) for 30 seconds. Any pouch that does not swell to a tightly distended form having at least one distorted edge while under the vacuum shall be classified as a major defect. After leakage testing, the envelopes shall be visually inspected for evidence of seal separation. Any seal separation of more than 1/16 inch from the food product edge of any seal shall be classified as a major defect. The lot size shall be

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expressed in units of pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.12 Pouch closure seal testing. The filled and sealed pouches shall be tested in accordance with ASTM Standard Test Method F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply tensile load to the specimen by movement of the upper or lower clamps may be used. The test specimens shall be cut to a length suitable for proper mounting. Three adjacent specimens, 1/2 or 1 inch wide shall be cut from the closure seal of each pouch in the sample. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. The results shall be reported to the nearest 0.1 pound per inch of width. Any specimen or average seal strength failing to meet the requirements of 5.1.1.4 shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The sample size shall be the number of pouches indicated by inspection level S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.13 Shipping container examination. The shipping containers shall be examined for the defects listed in table IV. The lot size shall be expressed in shipping containers. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects.

TABLE IV. Shipping container defects

Category	Defect
Major	
101	Metal fasteners on inside of box not covered by tape or paperboard
102	National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible
103	Net weight per shipping container exceeding 30 pounds

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5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. One hundred twenty five to 135 grams of instant oatmeal shall be filled into mixing bags.

5.1.1.1 Mixing bags. The mixing bags shall be fabricated from 0.003 inch thick, unpigmented, high density 0.941 to 0.965 g/cm^3 food grade, flat polyethylene tubing. The bottom seal shall be of adequate strength and integrity to withstand the addition of boiling water when tested in accordance with 4.5.3. The inside dimensions of the mixing bag shall be 7 inches wide by 8 1/2 inches long (±1/8 inch). The filled mixings bags shall be folded and placed into a pouch conforming to the requirements of 5.1.1.1.1.

5.1.1.2 Preformed pouch. The preformed pouch shall be fabricated from a multiple ply flexible laminate barrier material consisting of an outer ply of 0.0005 inch thick polyester bonded to an intermediate ply of 0.00035 inch thick aluminum foil with 10 pounds per ream pigmented polyethylene, and bonding the opposite side of the aluminum foil to 0.003 inch thick (inner ply) blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1, having a melt index range 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cm^3 in accordance with ASTM D 1505. As an alternative to the blend of linear low density polyethylene and polyethylene, an ionomer may be used. The pigmented polythylene shall be colored white overall with a color in the range of 37778 through 37886 of FED-STD-595. The material shall be suitably formulated for food packaging and shall not impart an odor or flavor to the product being packed. The material shall show no evidence of delamination or foreign odor when heat sealed into pouches. All thickness tolerances shall be ± 20 percent.

5.1.1.3 Pouch construction. The pouch shall be a flat style preformed pouch. The inside dimensions of the pouch shall be 4 5/8 inches wide by 8 1/4 inches long (± 1/8 inch). Pouches with smaller inside dimensions may be used provided the thickness of the filled and sealed pouch does not exceed 7/8 inch when examined in accordance with 4.5.6. The opening of the pouch between the heat sealed sides shall be the smaller dimension. The empty pouch shall be made by heat sealing three edges with 3/8 inch (± 1/8 inch) wide seals. The seals shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.5.4. A tear nick or notch at least 1/32 inch deep, located 1 to 1 1/4 inches from the top edge of the pouch shall be made in one or both side seals. The distance between the inside edge of the tear notch and the inside edge of the seal shall be at least 3/16 inch. One side of the open end of the pouch may be provided with an extended or foldover lip, extended not more than 5/16 inch to facilitate opening and filling. Tear notch location shall be measured from the top of the pouch, excluding the extended or foldover lip. Alternatively, if the pouch has serrated edges, the serrations may be used as tear notches, provided that the serrations are sharp (no plastic

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tailings exist) and the serration depth and the minimum seal width at the serrated edges are in accordance with the above requirements. The pouch shall be constructed with square or rounded corners at all four corners (radius approximately 3/8-inch).

5.1.1.4 Pouch filling and sealing. The filled and folded mixing bag shall be inserted into the pouches in such a manner as to avoid contaminating the heat seal area. The filled pouches shall be sealed, after manually evacuating excess air, with a 3/8 inch (±1/8 inch) heat seal. The sealed pouch shall show no leakage, delamination, or seal separation when examined as specified in 4.5.11. If thermal impulse closure seals are used, a seal width of 1/8 inch will be acceptable. The closure seal shall not extend below the tear notch on either side of the pouch. The closure seal shall be free of fold over wrinkles and delaminations. The closure seal shall have an average seal strength of not less than 6 pounds per inch and no individual specimen shall have a seal strength of less than 5 pounds per inch when tested as specified in 4.5.12. The closure seal shall be free of entrapped matter (for example, oats, powders, granules, mixing bag material, etc.) that reduces the effective closure seal to less than 1/16 inch wide.

5.2 Packing. Packing shall be level C.

5.2.1 Level C packing, for shipment to assembler. Not more than 30 pounds of one type of instant oatmeal shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the National Motor Freight Classification or Uniform Freight Classification as applicable, except fiberboard containers shall be closed in accordance with method II, as specified in the appendix of PPP-B-636. When metal fasteners are used in the manufacturers joint or setup of the fiberboard box, the fasteners on the inside of the box shall be covered with tape or paperboard to protect the pouches from damage.

5.3 Labeling and Marking.

5.3.1 Pouches. Each pouch shall be clearly printed or stamped, in a manner that does not damage the pouch, with permanent black ink which is free of carcinogenic elements or ingredients. All pouches shall be coded. If a non-contact type printer is used, the following information, as applicable, may be located anywhere on the pouch (in one complete print), except the closure seal area. In all cases, it is the contractor's responsibility to independently receive label approval from the USDA. The label shall include contractor's name, address, date markings, and weight. The following information shall be located on the body of the pouch not closer than 1/16 inch to any seal.

Product name and type (letters not less than 7/32 inch high)
Net weight (letters not less than 1/8 inch high)
ADD A MINIMUM OF 8 OUNCES OF HOT WATER (1/3 CANTEEN CUP).
SEAL WITH CLOSURE DEVICE. KNEAD TO MIX. WAIT ONE MINUTE BEFORE EATING.
OPTIONAL: CRUMBLED GRANOLA BAR, FRUIT SOUP, OR APPLE CIDER MAY BE ADDED FOR VARIETY. (letters not less than 5/32 inch high)

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Name and address of producer (letters not less than 3/32 inch high)
Date markings (Four digit Julian date code. The first digit of the code shall be the last digit of the current year, followed by the three digits representing the Julian calendar day- (for example, Jan. 1, 1989's code is 9001)-) (letters not less than 1/8 inch high)

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of product required (see 1.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article inspection. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Guar gum. The guar gum used in-house productions was guar gum 8/22 powder manufactured by TIC gums. Other guar gum meeting the requirements of 3.2.3 may be equally acceptable.

6.4 Apple flakes, low moisture. The apple flakes used in in-house production were low moisture apple flakes, code 183 and low moisture strawberry flavored and colored apple flakes with strawberry concentrate, code 188 manufactured by VACU-DRY. Other apple flakes meeting the requirements of 3.2.8 may be equally acceptable.

6.5 Flavoring, apple, natural and artificial. Acceptable alternatives to Flavor-Cap Natural and Artificial Apple Flavor #F22311 manufactured by McCormick-Stange Flavor Company, Inc. are Natural and Artificial Apple Flavor Flav-O-Lok 61087IU manufactured by Hercules Incorporated used at a 0.06 percent level and Natural and Artificial Fresh Apple Flavor Flav-O-Lok 695009U also manufactured by Hercules Incorporated used at a 0.04 percent level. Sugar may be used to adjust total ingredients to 100 percent. Other apple flavor meeting the requirements of 3.2.11 may be equally acceptable.

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6.6 Flavoring, strawberry, artificial. An acceptable alternative to Caps Artificial Strawberry Flavor No. 1979 manufactured by Borden is Artificial strawberry flavor #282-07-01/TP 05.00 manufactured by Firmenich used at the 0.1 percent level. Instant oats may be used to adjust ingredients to 100 percent. Other strawberry flavor meeting the requirements of 3.2.12 may be equally acceptable.

6.7 Subject term (key word) listing.

Breakfast cereal
Cereal

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No 8920-0523

Review activities:

Army - MD, TS
Navy - MC
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-44278A	Oatmeal, Instant (Operational Ration Component)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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